Year 10 Higher

REVISION PACK

Number 3

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 3 · Negative and fractional indices (Lessons 6.1, 6.2)

Q	Working	Answer
working out the value of powers		
1.1	**(a)** $3^{-2} = \frac{1}{3^2}$ **(b)** $\left(\sqrt[3]{27}\right)^2$ $= 3^2 = 9$ **(c)** $\left(\frac{27}{64}\right)^{\frac{1}{3}}$ $= \frac{3}{4}$	**(a)** $\frac{1}{9}$ **(b) 9** **(c)** $\frac{3}{4}$
1.2	**(a)** $2^{-5} = \frac{1}{2^5}$ **(b)** $\left(\sqrt{16}\right)^3$ $= 4^3 = 64$ **(c)** $\left(\frac{16}{9}\right)^{\frac{1}{2}}$ $= \frac{4}{3}$	**(a)** $\frac{1}{32}$ **(b) 64** **(c)** $\frac{4}{3}$
1.3	**(a)** $10^{-3} = \frac{1}{10^3}$ **(b)** $\left(\sqrt[3]{125}\right)^2$ $= 5^2 = 25$ **(c)** $\left(\frac{125}{8}\right)^{\frac{1}{3}}$ $= \frac{5}{2}$	**(a)** $\frac{1}{1000}$ **(b) 25** **(c)** $\frac{5}{2}$

1.4	**(a)** $4^{-2} = \frac{1}{4^2}$ **(b)** $\left(\sqrt{9}\right)^3$ $= 3^3 = 27$ **(c)** $\left(\frac{36}{25}\right)^{\frac{1}{2}}$ $= \frac{6}{5}$	**(a)** $\frac{1}{16}$ **(b)** **27** **(c)** $\frac{6}{5}$
1.5	**(a)** $5^{-3} = \frac{1}{5^3}$ **(b)** $\left(\sqrt[3]{8}\right)^4$ $= 2^4 = 16$ **(c)** $\left(\frac{64}{27}\right)^{\frac{1}{3}}$ $= \frac{4}{3}$	**(a)** $\frac{1}{125}$ **(b)** **16** **(c)** $\frac{4}{3}$
1.6	**(a)** $2^{-6} = \frac{1}{2^6}$ **(b)** $\left(\sqrt{100}\right)^3$ $= 10^3 = 1000$ **(c)** $\left(\frac{4}{49}\right)^{\frac{1}{2}}$ $= \frac{2}{7}$	**(a)** $\frac{1}{64}$ **(b)** **1000** **(c)** $\frac{2}{7}$
finding the power n		
1.7	**(a)** $\frac{1}{16} = \frac{1}{2^4} = 2^{-4}$ **(b)** $n - 3 = -1$	**(a)** $n = -4$ **(b)** $n = 2$
1.8	**(a)** $\frac{1}{27} = \frac{1}{3^3} = 3^{-3}$ **(b)** $n - 5 = -2$	**(a)** $n = -3$ **(b)** $n = 3$

1.9	**(a)** $\frac{1}{100} = \frac{1}{10^2} = 10^{-2}$ **(b)** $n - 7 = -3$	**(a)** $n = -2$ **(b)** $n = 4$
1.10	**(a)** $\frac{1}{64} = \frac{1}{4^3} = 4^{-3}$ **(b)** $n - 9 = -4$	**(a)** $n = -3$ **(b)** $n = 5$
1.11	**(a)** $\frac{1}{81} = \frac{1}{9^2} = 9^{-2}$ **(b)** $n - 3 = -2$	**(a)** $n = -2$ **(b)** $n = 1$
1.12	**(a)** $\frac{1}{8} = \frac{1}{2^3} = 2^{-3}$ **(b)** $n - 6 = -1$	**(a)** $n = -3$ **(b)** $n = 5$